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CUET UG

Accountancy

Computerised Accounting System

Official Syllabus Unit VI · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment & Exam Blueprint

Sub-topic	Official syllabus pointer (NTA CUET(UG) Accountancy 301, Unit VI)	Skills the questions test
Overview of CAS	Concept and types of computerised accounting system (CAS); Features of a CAS; Structure of a CAS; Software packages: Generic, Specific, Tailored	Classifying Ready-to-use, Customised, and Tailor-made software; identifying five AIS components.
Accounting & DBMS	Concept of entity and relationship: Designing database for accounting; Data hierarchy: character, field, record, file, database; Key concepts: Primary key, Foreign key	Applying relational data models; distinguishing Primary Keys and Foreign Keys; referential integrity.
Basic SQL for Accounting	Basic SQL queries to retrieve accounting data: SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY	Distinguishing DDL from DML; filtering with WHERE vs HAVING; computing aggregate totals.
Electronic Spreadsheets	Concept of electronic spreadsheet; Features offered by electronic spreadsheet; Cell referencing; Basic financial and accounting functions	Applying Relative, Absolute, and Mixed cell references; resolving spreadsheet error codes (#REF!, #DIV/0!).
Spreadsheet Applications	Applications of electronic spreadsheets in generating accounting information: Payroll accounting, Loan repayment schedule, Depreciation schedule	Calculating PMT, SLN, DB, and VLOOKUP; computing Net Salary from gross earnings and statutory deductions.

EXAM PATTERN

CUET UG Exam Blueprint: Unit VI contributes 4–6 questions (20–30 marks) in the 50-question paper. NTA frequently tests: (1) Sourcing criteria between Ready-to-Use, Customised, and Tailor-Made software, (2) Distinction between Primary Key and Foreign Key constraints, (3) Syntax and execution order of SQL queries (WHERE vs HAVING), (4) Difference between Absolute (`$A$1`) and Mixed (`$A1` or `A$1`) cell references, and (5) Financial function arguments in Excel (`PMT`, `SLN`, `DB`, `VLOOKUP`).

1. Overview of Computerised Accounting System (CAS)

A Computerised Accounting System (CAS) is an accounting information system that processes financial transactions and events according to Generally Accepted Accounting Principles (GAAP) to generate reports as per user requirements.

Core Operating Cycle of CAS

1. **Input:** Source documents (cash memos, invoices, vouchers) are digitized and keyed in through customized data entry screens.
2. **Processing:** The software processes transaction data against pre-defined accounting rules, automatically posting debits and credits into respective relational ledger tables.
3. **Output:** Real-time, instant generation of financial reports (Day Books, Ledgers, Trial Balance, Profit & Loss Account, Balance Sheet).

The Five Structural Components of an AIS

An Accounting Information System operates across five interrelated pillars:

Component	Technical Description	Accounting Application / Example
1. Hardware	Physical computing equipment and peripherals.	Desktop PCs, local servers, barcode scanners, laser printers for invoices.
2. Software	System software (OS) and application programs.	Operating system, RDBMS engine (SQL/Oracle), accounting packages (Tally, SAP).
3. People	Human users interacting with the system.	Data entry operators, accountants, internal auditors, Chief Financial Officer.
4. Procedures	Documented operational guidelines and rules.	Internal control manuals, chart of accounts guidelines, data backup policies.
5. Data	Raw facts and recorded transactions.	Voucher numbers, customer ledgers, account balances, inventory stock items.

Fundamental Features of CAS

- **Single-Point Data Entry:** Transaction data is keyed in only once via a digital voucher; all corresponding ledgers, trial balances, and financial statements update simultaneously in real time.
- **Speed and Accuracy:** Eliminates manual arithmetic errors, transposition errors, and tedious ledger balancing.
- **Online Real-Time Reporting:** Management can generate an instant balance sheet or cash position at any moment without waiting for year-end book closing.
- **Scalability and Storage:** Capable of handling massive transaction volumes across multi-branch and multi-currency operations.

2. Manual Accounting vs CAS & Codification Architecture

While both manual and computerised accounting systems adhere to fundamental double-entry principles, their execution mechanisms diverge sharply:

Comparative Matrix: Manual vs Computerised Accounting

Basis	Manual Accounting System	Computerised Accounting System (CAS)
Recording	Transactions recorded by hand in physical Journal Books.	Recorded digitally through electronic voucher entry screens.
Posting	Manual hand-posting from journals into ledger accounts.	Automatic, instantaneous posting executed by software.
Trial Balance	Prepared manually by extracting and totaling ledger balances.	Generated instantly on-demand; mathematically balanced.
Financial Reports	Prepared periodically (annually) after manual adjustments.	Available on-demand in real-time, reflecting live data.
Error Detection	Errors of casting, posting, and balancing are frequent.	Mathematical casting errors are eliminated by software validation.

Account Codification Schemes in CAS

In a CAS, account titles are assigned alphanumeric codes to expedite data entry and eliminate nomenclature ambiguity:

Coding Scheme	Design Principle	Concrete Accounting Example
Sequential Code	Consecutive numbers in unbroken sequence.	`001`, `002`, `003` (simple, but cannot insert new accounts).
Block Code	Dedicated numerical ranges for major heads.	`1000–1999` Assets, `2000–2999` Liabilities, `3000–3999` Capital.
Mnemonic Code	Alphabetic abbreviations aiding quick recall.	`FUR` (Furniture), `DEP` (Depreciation), `CHQ` (Cheque).
Hierarchical Code	Multi-tiered decimal parent-child hierarchy.	`1000` Current Assets → `1100` Bank → `1110` SBI Current A/c.

CUET CODIFICATION RULE

CUET Codification Rule: Block coding allows future expansion without disrupting existing numbers. Hierarchical coding provides infinite scalability for multi-branch corporate balance sheets.

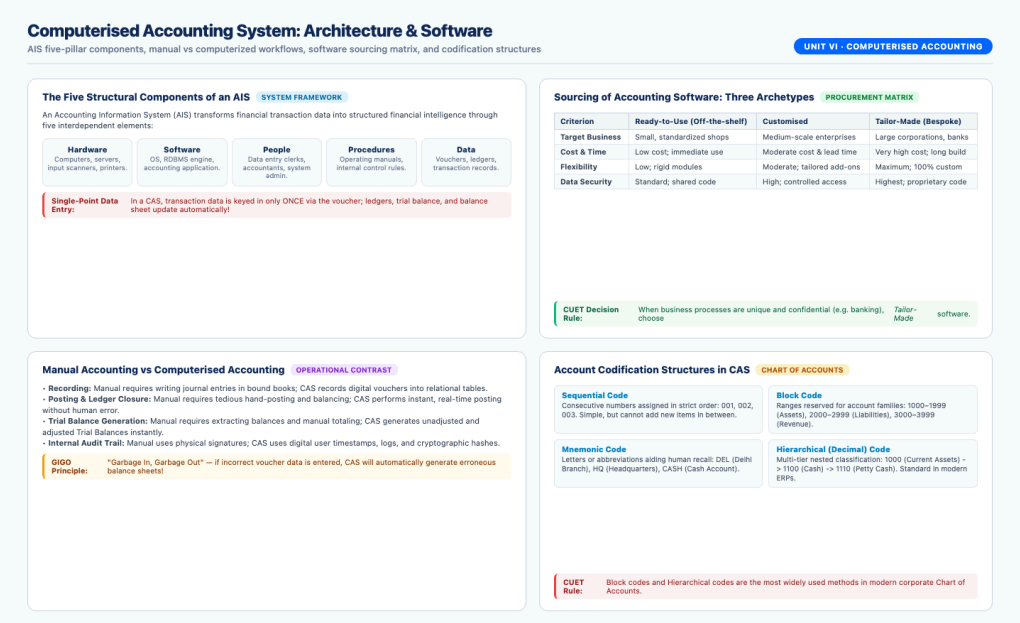


Figure 1: CAS Architecture: Five AIS pillars, manual vs computerized workflows, software sourcing matrix, and codification.

3. Sourcing & Selection of Accounting Software Packages

Sourcing Matrix: Ready-to-Use vs Customised vs Tailor-Made

Criterion	Ready-to-Use (Off-the-shelf)	Customised Software	Tailor-Made (Bespoke)
Target Users	Small retail shops, standard businesses.	Medium-scale enterprises.	Large corporations, banks, telecom.
Initial Cost	Very low; affordable off-the-shelf.	Moderate development cost.	Very high development & test cost.
Deployment Time	Immediate plug-and-play installation.	Requires moderate modification.	Long build, test, and release cycle.
Flexibility	Rigid; cannot modify core code.	Moderate; tailored add-ons.	100% custom-built to unique needs.
Data Security	Standard commercial security.	Enhanced access control.	Highest; proprietary encrypted code.
Staff Training	Minimal training required.	Moderate user training needed.	Specialised extensive training required.

Critical Factors Governing Accounting Software Selection

1. **Size and Scale of Business:** Small concerns thrive on off-the-shelf software; large multi-branch corporations mandate bespoke ERP suites.
2. **Cost of Installation & Maintenance:** Total ownership cost includes licensing, hardware upgrades, and recurring Annual Maintenance Contracts (AMC).
3. **Level of Secrecy & Data Security:** Highly sensitive financial records (banking, fintech) rule out generic packages in favour of tailor-made encrypted databases.
4. **Ease of Adaptation & Vendor Support:** Intuitive UI minimizes operator errors; reliable vendor support ensures timely statutory updates (e.g., GST changes).

4. Accounting and Database Management Systems (DBMS)

A Database Management System (DBMS) is software that enables structured storage, retrieval, and modification of financial data across linked relational tables.

Hierarchy of Accounting Data

Accounting data within an RDBMS is structured hierarchically from atomic bits to full databases:

- **Bit & Byte:** A bit is the basic binary unit (0 or 1); 8 bits form a **Byte/Character** (e.g., 'A', '₹').
- **Field / Attribute (Column):** A distinct accounting property (e.g., `Voucher_No`, `Acc_Name`, `Amount`).
- **Record / Tuple (Row):** A complete collection of fields representing one transaction instance.
- **Table / Relation (File):** A collection of logically related records sharing identical attributes.
- **Database:** The master integrated repository comprising all interrelated accounting tables.

Relational Database Keys Architecture

Relational databases enforce key constraints to guarantee entity integrity and eliminate orphan records:

Key Type	Technical Definition	Accounting Application / Example
Primary Key	Uniquely identifies every record in a table. Must be unique and **cannot contain NULL values**.	`Voucher_ID` in Voucher Table; `Acc_Code` in Accounts Master Table.
Foreign Key	Attribute referencing the Primary Key of another table, linking parent and child entities.	`Acc_Code` inside Voucher Table, referencing `Acc_Code` in Accounts Master.
Candidate Key	Any minimal set of attributes capable of uniquely identifying table records.	`PAN_Number` or `Bank_Account_No` in Vendor or Employee Master.
Alternate Key	A Candidate Key not chosen as the Primary Key.	If `Acc_Code` is Primary Key, unique `Acc_Name` is an Alternate Key.

REFERENTIAL INTEGRITY CONSTRAINT

Referential Integrity mandates that a Foreign Key in a child table (e.g., Voucher Table) MUST match an existing Primary Key in the parent master table (e.g., Accounts Master). A user **cannot post a voucher** to an account code that does not exist in the master table!

5. Basic SQL Queries for Accounting

Structured Query Language (SQL) is the standard language used to define, manipulate, and query data in a relational database.

Classification of SQL Commands

- **DDL (Data Definition Language):** Used to build, alter, and delete database structures:
  - `CREATE TABLE`: Defines table schema, column data types, and primary key constraints.
  - `ALTER TABLE`: Modifies an existing table (e.g., adds a new GST column).
  - `DROP TABLE`: Permanently deletes an entire table and its schema from the database.
- **DML (Data Manipulation Language):** Used to query, insert, update, and manage table records:
  - `SELECT`: Retrieves data from tables based on specified query criteria.
  - `INSERT INTO`: Adds new transaction rows into an accounting table.
  - `UPDATE`: Modifies existing field values (e.g., updates vendor credit limit).
  - `DELETE`: Removes specific transaction records from a table.

Anatomy of an Accounting SQL SELECT Query

```
SELECT Column_1, Aggregate_Function(Column_2)
FROM Table_Name
WHERE Row_Condition
GROUP BY Grouping_Column
HAVING Group_Condition
ORDER BY Sorting_Column ASC|DESC;
```

Clause	Function in Accounting Query	Concrete Accounting Example
SELECT	Specifies attributes to retrieve or aggregate.	`SELECT Acc_Name, SUM(Debit_Amount)`
FROM	Specifies source database table.	`FROM Daily_Vouchers`
WHERE	Filters individual raw rows before grouping.	`WHERE Voucher_Date >= '2024-04-01'`
GROUP BY	Collapses rows sharing identical values into groups.	`GROUP BY Acc_Name`
HAVING	Filters aggregated groups after `GROUP BY`.	`HAVING SUM(Debit_Amount) > 100000`
ORDER BY	Sorts final output ascending (ASC) or descending.	`ORDER BY SUM(Debit_Amount) DESC`

CUET HIGH-FREQUENCY SQL TRAP: WHERE VS HAVING

- `WHERE` clause filters **individual records** before aggregation and CANNOT contain aggregate functions like `SUM()` or `COUNT()`.
- `HAVING` clause filters **aggregated groups** and is used exclusively after `GROUP BY`.

6. Electronic Spreadsheets in Accounting

An electronic spreadsheet (e.g., Microsoft Excel, Google Sheets) is an interactive grid of rows and columns used to manipulate, calculate, and model financial data.

Cell Referencing Archetypes

Cell referencing defines how cell addresses behave when formulas are copied across rows or columns:

1. **Relative Referencing (`A1`):**
   - The cell address shifts dynamically relative to the direction of copying.
   - **Example:** If `=A1+B1` in cell `C1` is copied down to cell `C2`, it automatically shifts to `=A2+B2`.
2. **Absolute Referencing (`$A$1`):**
   - Both column and row are locked with dollar signs (`$`). The reference remains completely fixed regardless of where the formula is copied.
   - **Example:** If `=A1*$D$1` is copied from `B1` to `B2`, it becomes `=A2*$D$1` (the tax rate in `$D$1` stays locked).
3. **Mixed Referencing (`$A1` or `A$1`):**
   - Either the column is locked (`$A1`) while the row remains free, or the row is locked (`A$1`) while the column remains free.
   - **Example:** In a two-way multiplication table, `=$A2*B$1` locks column A for rates and row 1 for quantities.

Common Spreadsheet Error Values in Accounting

Error Code	Underlying Cause	Practical Accounting Remedy
`\#DIV/0!`	Formula attempts division by zero or an empty cell.	Ensure denominator cell contains non-zero value or wrap in `IF()` check.
`\#N/A`	Lookup value is not available in lookup table.	Check `VLOOKUP` search key; verify table array range covers search key.
`\#VALUE!`	Incorrect data type passed to function (e.g., text in math).	Ensure cells referenced in arithmetic contain numeric currency data.
`\#REF!`	Cell reference is invalid (referenced cell was deleted).	Re-enter formula; avoid deleting source rows/columns used in calculations.
`\#NAME?`	Unrecognized function name or misspelled formula.	Correct formula spelling (e.g., `SUMM` corrected to `SUM`).
`\#\#\#\#\#`	Column width is too narrow to display full numeric value.	Double-click column header boundary to auto-expand cell width.

7. Financial, Logical & Lookup Functions

Modern spreadsheets provide specialized built-in functions for financial modeling, depreciation calculation, and data retrieval.

Key Financial Functions

1. **`PMT(rate, nper, pv, [fv], [type])`:**
   - Computes the fixed periodic payment (EMI) for a loan based on constant payments and constant interest rate.
   - **Rate:** Interest rate per period. (For monthly EMI at 12% p.a., Rate $= 12\frac{\%}{12} = 1\%$ or 0.01).
   - **Nper:** Total number of payment periods. (For a 5-year loan paid monthly, Nper $= 5 \times 12 = 60$).
   - **Pv:** Present value or principal loan amount borrowed.
2. **`PV(rate, nper, pmt, [fv], [type])`:** Calculates the present lump-sum value of a series of future cash flows.
3. **`FV(rate, nper, pmt, [pv], [type])`:** Calculates the future compounded value of an investment.
4. **`RATE(nper, pmt, pv, [fv])`:** Calculates the periodic interest rate of an annuity or loan.
5. **`NPER(rate, pmt, pv, [fv])`:** Calculates the total number of payment periods required to clear a loan.

Depreciation Functions

- **`SLN(cost, salvage, life)` — Straight-Line Method:**

$$\text{Annual Depreciation} = \frac{\text{Cost} - \text{Salvage Value}}{\text{Useful Life}}$$

  - Computes equal, constant depreciation for each period.
- **`DB(cost, salvage, life, period, [month])` — Declining Balance Method:**
  - Computes depreciation at a fixed declining percentage rate (Written Down Value method) for a specific individual period.

Logical and Lookup Functions

- **`IF(logical_test, value_if_true, value_if_false)`:**
  - Evaluates a condition and returns one value if TRUE and another if FALSE.
  - **Accounting Example:** `=IF(E2>=50000, E2*10%, E2*5%)` (calculates sales bonus).
- **`VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`:**
  - Searches vertically down the leftmost column of `table_array` for `lookup_value` and returns a value from `col_index_num`.
  - Set `range_lookup = FALSE` (or `0`) for an **exact match** (mandatory for account codes and employee IDs!).

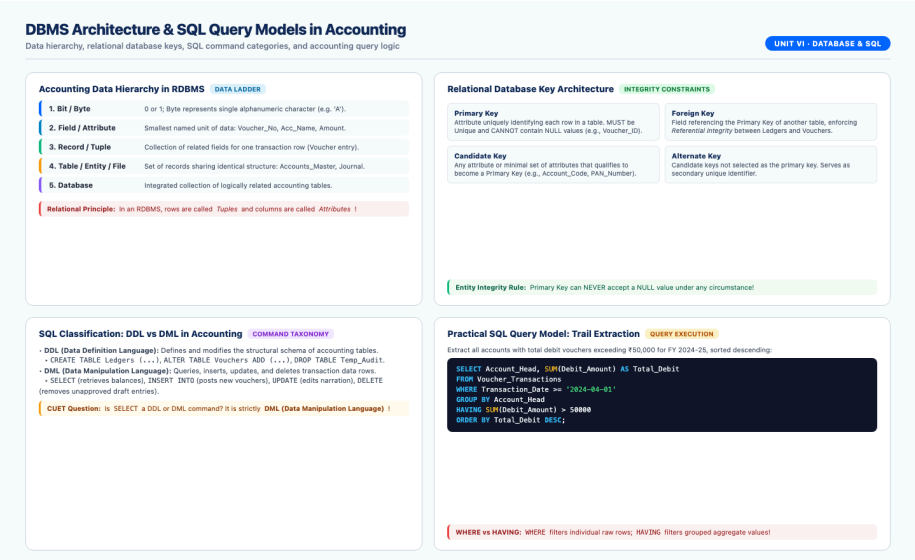


Figure 2: Relational Database & SQL Architecture: DBMS schema, data hierarchy, and core SQL queries.

8. Payroll Accounting & Financial Modeling in Spreadsheets

Payroll accounting involves calculating gross earnings, statutory deductions, and net salary for all employees on a recurring periodic schedule.

Core Components of a Payroll Schedule

- **Gross Pay:** Basic Pay + Dearness Allowance (DA) + House Rent Allowance (HRA) + Conveyance/Transport Allowance.
- **Statutory Deductions:**
  - **Employees' Provident Fund (EPF):** 12% of (Basic Pay + DA).
  - **Employee State Insurance (ESI):** Statutory percentage of Gross Salary for eligible salary thresholds.
  - **Professional Tax:** Fixed state-specific statutory levy (e.g., ₹200/month).
  - **Tax Deducted at Source (TDS):** Progressive income tax deduction based on projected annual tax liability.
- **Net Salary Formula:**

$$\text{Net Salary} = \text{Gross Earnings} - \text{Total Deductions}$$

MASTER PROBLEM — LOAN AMORTISATION & DEPRECIATION MODELING

1. **Loan EMI Computation:** A company takes a bank loan of ₹6,00,000 at 12% p.a. for 3 years repayable monthly.
   - Monthly Rate = 12% / 12 = 1% = 0.01; Nper = 3 × 12 = 36 months; Pv = ₹6,00,000.
   - Spreadsheet formula: `=PMT(0.01, 36, -600000)` yields **₹19,929** per month.
2. **Straight-Line Depreciation:** Plant purchased for ₹5,00,000; Salvage value ₹50,000; Useful life 5 years.
   - Spreadsheet formula: `=SLN(500000, 50000, 5)` yields (₹5,00,000 − ₹50,000) / 5 = **₹90,000** per year.

9. Security, Internal Controls & Audit Trail in CAS

Information security in computerised accounting ensures confidentiality, integrity, and availability of financial records.

Threats to Accounting Information Systems

- **Internal Threats:** Unauthorized data alteration, employee fraud, accidental erasure, password sharing.
- **External Threats:** Malware, ransomware infections, phishing attacks, unauthorized remote intrusions.

Multi-tier Access Control Architecture

1. **Authentication Controls:** Multi-factor authentication (MFA), biometric verification, complex passwords.
2. **Authorization & Role-Based Access Control (RBAC):** Restricting voucher entry, posting, and reporting privileges by user role.
3. **Data Encryption:** Cryptographic encoding of database files in transit and at rest.

Digital Audit Trail Requirement (MCA Mandate)

Under the Companies (Accounts) Rules, accounting software used by registered companies must maintain an immutable, unalterable digital audit trail:

- Automatically logs every transaction creation, modification, and deletion.
- Captures exact date, time, and user ID timestamps for all edits.
- The audit trail feature **cannot be disabled or bypassed**.

Data Backup and Recovery Protocol

- **Grandfather-Father-Son (GFS) Rotation:** Systematic daily, weekly, and monthly offsite backups.
- **Disaster Recovery Plan (DRP):** Redundant cloud synchronization to maintain business continuity.

MASTER PROBLEM 1 — AUTOMATED PAYROLL ACCOUNTING SCHEDULE

Fact Pattern: Excel Enterprises compensation rules: Basic Pay: Amit ₹60,000, Priya ₹50,000, Rohit ₹40,000. DA = 40% of Basic; HRA = 20% of Basic; Conveyance = ₹3,000 fixed. Deductions: PF = 12% of (Basic + DA); Prof. Tax = ₹200; TDS = 10% of Gross.

- **Amit Calculations:** Gross = 60,000 + 24,000 + 12,000 + 3,000 = **₹99,000**. Deductions: PF (12% × ₹84,000 = ₹10,080) + Tax ₹200 + TDS (10% × ₹99,000 = ₹9,900) = **₹20,180**. Net Salary = **₹78,820**.

Employee	Basic (₹)	DA (₹)	HRA (₹)	Gross (₹)	Deductions (₹)	Net Salary (₹)
Amit	60,000	24,000	12,000	99,000	20,180	**78,820**
Priya	50,000	20,000	10,000	83,000	16,900	**66,100**
Rohit	40,000	16,000	8,000	67,000	13,620	**53,380**
Total	**1,50,000**	**60,000**	**30,000**	**2,49,000**	**50,700**	**1,98,300**

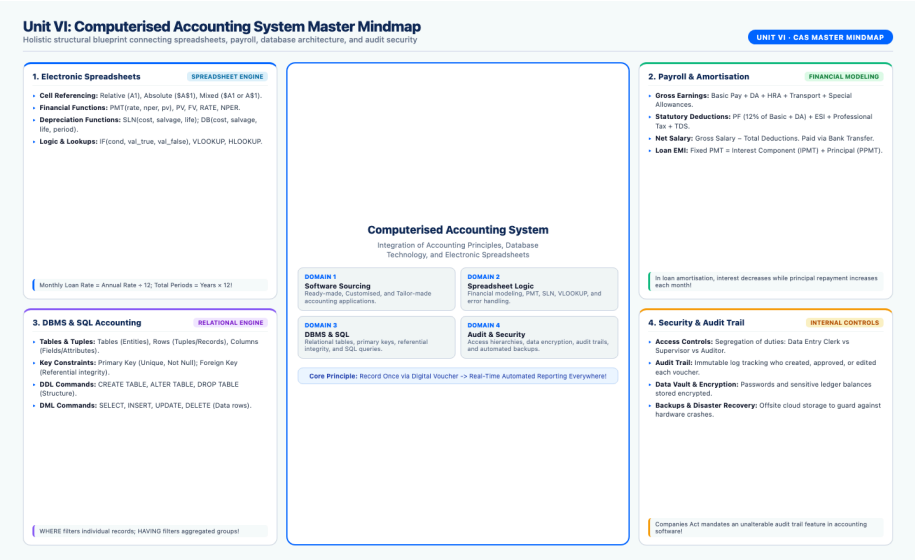


Figure 3: Master Synthesis Mindmap: Structural framework of Unit VI Computerised Accounting System.

11. Strategic Examination Synthesis

The complete computerised accounting architecture connects four foundational domains:

1. **System Sourcing Decision:** Balancing business scale, customization needs, and data security between Ready-to-Use, Customised, and Tailor-Made software.
2. **RDBMS Backbone:** Managing transaction records across tables linked by Primary Key and Foreign Key constraints, enforced through referential integrity.
3. **Spreadsheet Engine:** Leveraging financial (`PMT`, `PV`), depreciation (`SLN`, `DB`), and lookup (`VLOOKUP`) formulas to automate payroll, amortisation, and analytical reporting.
4. **Internal Controls & Audit:** Protecting corporate accounting data through access segregation, encrypted vaults, and immutable digital audit trails.

MASTER COMPUTERISED ACCOUNTING REVISION CHEATSHEET

- **AIS Five Pillars:** Hardware, Software, People, Procedures, Data. Transaction entered once via digital voucher!
- **Software Sourcing:** Ready-to-use (Low cost, rigid) | Customised (Moderate cost) | Tailor-made (High cost, bespoke).
- **Data Hierarchy:** Bit -> Byte -> Field (Attribute/Column) -> Record (Tuple/Row) -> Table -> Database.
- **Key Constraints:** Primary Key (Unique, Non-Null) | Foreign Key (Referential Integrity to parent table).
- **SQL Taxonomy:** DDL: `CREATE`, `ALTER`, `DROP` | DML: `SELECT`, `INSERT`, `UPDATE`, `DELETE`.
- **WHERE vs HAVING:** `WHERE` filters individual records; `HAVING` filters aggregated groups after `GROUP BY`.
- **Cell Referencing:** Relative (`A1`), Absolute (`$A$1`), Mixed (`$A1` or `A$1`).
- **Financial Formulas:** `PMT(rate, nper, pv)` | Rate = Annual Rate / 12 | Nper = Years × 12.
- **Depreciation Functions:** `SLN(cost, salvage, life)` = Straight-Line | `DB(cost, salvage, life, period)` = WDV.
- **Payroll Equations:** Gross = Basic + DA + HRA + Allowances | PF = 12% of (Basic + DA) | Net = Gross − Deductions.

TOP 7 HIGH-FREQUENCY CUET TRAPS IN COMPUTERISED ACCOUNTING

1. **SELECT Command Classification:** `SELECT` is strictly a **DML (Data Manipulation Language)** command, NOT DDL!
2. **Primary Key NULL Trap:** A Primary Key can NEVER contain a NULL value under any circumstances.
3. **Monthly Rate & Period Trap in PMT:** Never enter annual rate directly in `PMT` for monthly EMI; divide by 12 and multiply years by 12!
4. **WHERE vs HAVING Aggregate Trap:** You cannot use `SUM()` or `COUNT()` inside a `WHERE` clause; use `HAVING`!
5. **VLOOKUP Exact Match Flag:** Always specify `FALSE` (or `0`) as the 4th argument in `VLOOKUP` to prevent erroneous approximate matches!
6. **Provident Fund Base Trap:** PF is calculated on **(Basic Pay + DA)**, NOT on Gross Salary!
7. **Audit Trail Feature Mandate:** Companies Act mandates an unalterable audit trail feature; software without it is legally non-compliant.

CUET UG Exam-Style Practice Questions

Q1. Which of the following is NOT a component of an Accounting Information System (AIS)?

(A) Hardware
(B) Procedures and Instructions
(C) Data and People
(D) Physical Cash Inventory

Sol. Q1. An AIS consists of five structural components: Hardware, Software, People, Procedures, and Data. Physical cash is a current asset, not an architectural component of the information system.

Answer: (D)

Q2. A large nationalized bank requiring unique, confidential, and multi-branch transaction software should ideally opt for which type of software?

(A) Tailor-Made (Bespoke) Software
(B) Ready-to-Use Software
(C) Generic Spreadsheet Software
(D) Off-the-shelf Free Software

Sol. Q2. Large corporations and banks with specialized, high-volume, and confidential transactions require tailor-made (bespoke) software developed specifically to match their unique operational and security needs.

Answer: (A)

Q3. In a Relational Database Management System (RDBMS), the rows and columns of a table are formally termed as:

(A) Tuples and Attributes
(B) Fields and Records
(C) Keys and Domains
(D) Entities and Schemas

Sol. Q3. In formal relational database theory, rows representing individual records are called **Tuples**, and columns representing specific data characteristics are called **Attributes**.

Answer: (A)

Q4. Which of the following constraints guarantees that an attribute in a child table must match an existing primary key value in a parent master table?

(A) Domain Integrity
(B) Entity Integrity
(C) Referential Integrity
(D) Unique Key Integrity

Sol. Q4. Referential integrity is enforced through Foreign Keys, ensuring that relationships between tables remain consistent and preventing orphan records in transaction tables.

Answer: (C)

Q5. Which of the following SQL commands belongs to the category of Data Definition Language (DDL)?

(A) SELECT
(B) INSERT
(C) CREATE
(D) UPDATE

Sol. Q5. `CREATE`, `ALTER`, and `DROP` are DDL commands because they define or modify database schemas. `SELECT`, `INSERT`, `UPDATE`, and `DELETE` are DML commands.

Answer: (C)

Q6. In an SQL query, which clause is used to filter records AFTER an aggregate function has been applied?

(A) HAVING (B) WHERE
(C) ORDER BY (D) GROUP BY

Sol. Q6. The `HAVING` clause is used to filter aggregated group results created by `GROUP BY`. In contrast, the `WHERE` clause filters individual records before any grouping or aggregation takes place.

Answer: (A)

Q7. If the formula `=$A$1*B1` located in cell `C1` is copied and pasted into cell `C2`, what formula will appear in cell `C2`?

(A) `=$A$1*B2` (B) `=$A$2*B2`
(C) `=A1*B2` (D) `=$A$1*$B$1`

Sol. Q7. `$A$1` is an absolute reference and remains fixed. `B1` is a relative reference and shifts down one row to `B2`. Thus, the copied formula becomes `=$A$1*B2`.

Answer: (A)

Q8. Which spreadsheet error code indicates that a cell reference in a formula is invalid because the referenced cell was deleted?

(A) `\#DIV/0!` (B) `\#VALUE!`
(C) `\#REF!` (D) `\#NAME?`

Sol. Q8. The `\#REF!` error occurs when a formula refers to a cell that is not valid, commonly caused when referenced cells, rows, or columns are deleted or overwritten.

Answer: (C)

Q9. A machine is purchased for ₹2,00,000 with an estimated salvage value of ₹20,000 and a useful life of 5 years. What is the annual depreciation returned by the spreadsheet formula `=SLN(200000, 20000, 5)`?

(A) ₹40,000 (B) ₹30,000
(C) ₹44,000 (D) ₹36,000

Sol. Q9. `SLN` computes Straight-Line Depreciation: (Cost − Salvage) / Life = (₹2,00,000 − ₹20,000) / 5 = ₹1,80,000 / 5 = **₹36,000**.

Answer: (D)

Q10. When using the `VLOOKUP` function to retrieve an exact ledger balance matching an account code, what value should be passed as the fourth argument (`[range_lookup]`)?

(A) `TRUE` or `1` (B) `"APPROX"`
(C) `FALSE` or `0` (D) `OMITTED`

Sol. Q10. Setting the fourth argument to `FALSE` (or `0`) forces `VLOOKUP` to find an exact match for the lookup key, preventing erroneous approximate data retrieval.

Answer: (C)

Q11. An employee has a Basic Pay of ₹40,000 and Dearness Allowance (DA) of ₹16,000. Statutory Provident Fund (PF) deduction @ 12% amounts to:

(A) ₹4,800 (B) ₹5,600
(C) ₹1,920 (D) ₹6,720

Sol. Q11. Under statutory payroll rules, Provident Fund is calculated at 12% on (Basic Pay + DA): 12% × (₹40,000 + ₹16,000) = 12% × ₹56,000 = **₹6,720**.

Answer: (D)

Q12. A company takes a 5-year loan of ₹10,00,000 at 12% per annum, repayable in monthly instalments. In the formula `=PMT(rate, nper, pv)`, what values should be passed for `rate` and `nper`?

(A) 12% and 5 (B) 1% and 60
(C) 12% and 60 (D) 1% and 5

Sol. Q12. For monthly payments: Rate = 12% / 12 = 1% (or 0.01), and Nper = 5 × 12 = 60 monthly periods.

Answer: (B)

Q13. In an account codification scheme, reserving the range 1000–1999 for Assets and 2000–2999 for Liabilities is an example of:

(A) Sequential Code (B) Block Code
(C) Mnemonic Code (D) Alphabetical Code

Sol. Q13. A Block Code assigns a distinct block or range of numbers to specific classifications or account families, allowing convenient grouping and room for future additions.

Answer: (B)

Q14. Which of the following internal control features is MANDATED by corporate law in accounting software used by registered companies in India?

(A) Automatic cloud file sharing (B) Unalterable audit trail recording an edit log of all changes
(C) Direct integration with social media platforms (D) Voice recognition voucher entry

Sol. Q14. The Ministry of Corporate Affairs mandates that companies must use accounting software equipped with an unalterable audit trail feature to record every transaction change with date and user timestamps.

Answer: (B)

Q15. Which of the following describes the principle of 'GIGO' in computerised accounting?

(A) General Ledger In, General Ledger Out (B) Garbage In, Garbage Out
(C) Good Input, Guaranteed Output (D) Graphic Interface Graphical Output

Sol. Q15. 'Garbage In, Garbage Out' (GIGO) states that the accuracy of computerised accounting reports is entirely dependent on the quality of input data; erroneous input vouchers inevitably yield erroneous financial statements.

Answer: (B)